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Dear Hiring Manager,

I am writing to apply for a research internship in distributed systems with your team, between May and October 2027. I am a computer science engineering student at CESI in France, where I maintain a 3.8/4.0 GPA, and I have spent the last two years working as an apprentice software engineer on a production codebase. I am looking for an internship where I can work on open questions rather than known features, and learn from researchers how to turn an intuition into a result that holds up.

At R3mScore, I build and ship features that customers use the same week. I designed and delivered a Word-to-survey import and export engine on my own, from requirements to production. It parses .docx questionnaires into structured surveys with conditional logic, loops and adaptive questions, and it replaces up to three hours of manual setup per survey. I also migrated a 15,000-line frontend from JavaScript to TypeScript incrementally, with no feature freeze and no regression in production, and added JUnit and Vitest coverage on the critical paths. Working this way taught me to keep a codebase releasable at every commit and to know which parts of a system I have not yet proven correct.

My projects are where I go after the problems a product codebase does not hand me. As tech lead on a team of four, I architected an eight-service distributed platform with gRPC and RabbitMQ messaging, one database per service behind an Nginx gateway, and distributed tracing through OpenTelemetry. I wanted to see where a distributed system actually breaks: partial failure, retries, and the gap between what a trace shows and what a log shows. On a team of five, I also implemented Tabu Search and multi-start Simulated Annealing for a constrained Traveling Salesman Problem, and built the statistical framework comparing our algorithms from 5 to 3,000 nodes over multiple random seeds. It is where I learned that a result only counts once it survives a change of seed and of scale.

What I am looking for is a research team working on distributed systems, whose researchers will tell me when my first design or my first experiment is wrong. My day-to-day stack is Java and Spring, with SQL beneath it and TypeScript and React on top, and I use Python and C# on school and personal projects, but I care less about which stack I land in than about working on problems where performance, reliability and correctness have to be measured rather than assumed. One practical point I would rather raise early: my school sets my internship window, which runs from 3 May to 15 October 2027, and the internship must last at least 17 weeks within it, in a research laboratory or an R&D team under the supervision of an engineer or researcher.

I would bring two years of production experience, a habit of owning features end to end, and the willingness to ask questions until I actually understand the system I am changing. My resume is attached, and I would be glad to discuss how my background fits your team, or how the programme dates could work. Please feel free to reach out by email at contact@clementomnes.dev or by phone at +33 X XX XX XX XX. Thank you for your time and consideration.

Sincerely,

Clément Omnès